Oscillations in a charged cylinder — Solution

X22 (2022) — English translation

A1 **0.50 pts**

A disk with radius R is charged with surface charge density σ_R. Determine the potential $\varphi(y)$ at a point on the axis at a distance y from the center of the disk. The potential is zero at infinity.

Potential from a ring of width dr with radius r

$$d\varphi = k\frac{\sigma_R \cdot 2\pi r dr}{\sqrt{y^2+r^2}}$$

Potential от всего диска we find интегрированием:

$$\varphi(y) = \pi k\sigma_R \int_0^R \frac{d(r^2)}{\sqrt{y^2+r^2}} = 2k\sigma_R\left(\sqrt{R^2+y^2}-|y|\right)$$

Answer: \textbfAnswer:

$$\varphi(y) = 2k\sigma_R\left(\sqrt{R^2+y^2}-|y|\right)$$

A2 **1.00 pts**

Two such disks of radius R charged with surface charge density $\sigma_R > 0$ are parallel to each other. The distance between the centers of the disks is $2L$, the centers are located on the axis of the disks. In the equilibrium position there is a charge q with mass m, which can only move along the axis of the disks. Determine the angular frequency ω_1 of vibrations of such a charge. What is the charge sign?

When the charge is displaced by x along the axis, an outer ring of width $\delta = 2Rx/L$ appears on the nearest disk, which creates a restoring force. The total field of the remaining charges is zero. The field of this ring is

$$E_x(x) = -k\frac{2\pi R\delta\sigma_R}{(L^2+R^2)}\frac{L}{\sqrt{L^2+R^2}} = -\frac{\sigma_R R^2}{\varepsilon_0(L^2+R^2)^{3/2}}x.$$

Equation движения chargea

$$m\ddot{x} = qE(x).$$

Angular frequency for/at колебаниях:

$$\omega^2 = \frac{q\sigma_R R^2}{m\varepsilon_0(L^2+R^2)^{3/2}}.$$

Answer: \textbfAnswer:

$$\omega^2 = \frac{q\sigma_R R^2}{m\varepsilon_0(L^2+R^2)^{3/2}},$$

$$q > 0.$$

A3 — 1.00 pts

Now this charge can only move in a perpendicular direction. Express the angular frequency ω_2 of oscillations in this case in terms of ω_1. What is the charge sign now?

We use Gauss's theorem for a small cylinder between the disks to find the dependence of the radial field $E_r(r)$ on the displacement from the axis r:

$$2\pi r \cdot 2xE_r(r) + 2\pi r^2 E_x(x) = 0$$

$$E_r(r) = -\frac{1}{2}E_x(x)\frac{r}{x}.$$

Charge должен быть другого signa. It can be seen that coefficient в линейной зависимости в 2 са меньше, therefore,

$$\omega_2 = \frac{\omega_1}{\sqrt{2}}.$$

Answer: \textbfAnswer:

$$\omega_2 = \frac{\omega_1}{\sqrt{2}},$$

$$q < 0.$$

B1 — 1.00 pts

The lateral surface of a cylinder with radius R and length L is charged with surface charge density σ_L. Determine the potential at a point on the axis at a distance z from the center of one of the bases of the cylinder. The potential is zero at infinity.

Potential from a ring of height dl at a distance l from zero

$$d\varphi(y) = k\frac{\sigma_L \cdot 2\pi R dl}{\sqrt{(z+l)^2 + R^2}}.$$

Potential от всей боковой поверхности cylindera we find интегрированием:

$$\varphi(y) = 2\pi k R\sigma_L \int_z^{L+z} \frac{d(l+z)}{\sqrt{(l+z)^2+R^2}} = 2\pi k R\sigma_L \left(\text{arth}\frac{L+z}{\sqrt{(L+z)^2+R^2}} - \text{arth}\frac{z}{\sqrt{z^2+R^2}}\right).$$

Answer: \textbfAnswer:

$$\varphi(y) = 2\pi k R\sigma_L \left(\text{arth}\frac{L+z}{\sqrt{(L+z)^2+R^2}} - \text{arth}\frac{z}{\sqrt{z^2+R^2}}\right).$$

B2 — 1.00 pts

Two such cylinders (with radius R and length L, surface charged with surface charge density $\sigma_L > 0$) are placed close together and have a common axis. In the equilibrium position there is a charge q with mass m, which can only move along the axis of the cylinders. Determine the angular frequency ω_3 of vibrations of such a charge. What is the charge sign?

When the charge is displaced by x along the axis, an outer ring of width $\delta = 2x$ appears on the far cylinder, which creates a restoring force. The total field of the remaining charges is zero. The field of this ring is

$$E_x(x) = k\frac{2\pi R\delta\sigma_L}{(L^2+R^2)}\frac{L}{\sqrt{L^2+R^2}} = \frac{\sigma_L RL}{\varepsilon_0(L^2+R^2)^{3/2}}x.$$

Equation движения chargea

$$m\ddot{x} = qE_x(x),$$

it can be seen that charge должен быть отрицательным. Angular frequency for/at колебаниях:

$$\omega_3^2 = \frac{|q|\sigma_L RL}{m\varepsilon_0(L^2+R^2)^{3/2}}.$$

Answer: \textbfAnswer:

$$\omega_3^2 = \frac{|q|\sigma_L RL}{m\varepsilon_0(L^2+R^2)^{3/2}},$$
$$q < 0.$$

B3 — 0.50 pts

Now this charge can only move in a perpendicular direction. Express the angular frequency ω_4 of oscillations in this case in terms of ω_3. What is the charge sign now?

Everything happens similarly to A3, that is,

$$E_r(r) = -\frac{1}{2}E_x(x)\frac{r}{x},$$

therefore $\omega_4^2 = \omega_3^2/2$ and the charge should also change sign.

Answer: \textbfAnswer:

$$\omega_4 = \frac{\omega_3}{\sqrt{2}},$$
$$q > 0.$$

C1 — 1.50 pts

A charged cylinder with radius R and height $L = 40R/9$ consists of a side surface and one base. Surface charge density of the side surface σ_L, base σ_R. If you place a point charge at the center of the opposite base, it will be in an equilibrium position. Define the relation σ_L/σ_R.

The field strength created by the base of the cylinder will be found as

$$E_R = -\left.\frac{d\varphi}{dy}\right|_L.$$

Эта напряженность equals

$$E_R = 2\pi\sigma_R k\left(1 - \frac{L}{\sqrt{R^2+L^2}}\right).$$

Напряженность поля, создаваемого боковыми стенками cylindera let us find как

$$E_L = -\left.\frac{d\varphi}{dz}\right|_0.$$

Эта напряженность equals

$$E_L = 2\pi\sigma_L k\left(1 - \frac{R}{\sqrt{R^2+L^2}}\right).$$

Суммарное поле equals нулю, hence we get:

$$\frac{\sigma_L}{\sigma_R} = -\frac{\sqrt{R^2+L^2}-L}{\sqrt{R^2+L^2}-R} = -\frac{41-40}{41-9} = -\frac{1}{32}$$

Answer: \textbfAnswer:

$$\frac{\sigma_L}{\sigma_R} = -\frac{\sqrt{R^2+L^2}-L}{\sqrt{R^2+L^2}-R} = -\frac{1}{32}$$

C2 **2.50 pts**

A charged cylinder with radius $R = 28b$ and height $L = 45b$ consists of a side surface and one base. Side surface charge $\sigma_L = -8\sigma_0$, base charge $\sigma_R = 25\sigma_0 > 0$. A particle with charge $q > 0$ is placed on the axis of this system. Estimate numerically the z coordinates (in b units) of the equilibrium positions if the particle can only move along the axis. Coordinate z is calculated as in the picture. Do this as accurately as possible, however, 1% accuracy is sufficient. Answers falling within 1% of correct will receive full marks.

We are looking for a point at which the total field is equal to zero. Next, we measure all lengths in units of b. For points with coordinates $z > -45$ we obtain the equation

$$E(z) + E(z) = 0,$$

$$25\left(\frac{45+z}{\sqrt{28^2+(45+z)^2}} - 1\right) - 8\left(\frac{28}{\sqrt{28^2+(45+z)^2}} - \frac{28}{\sqrt{28^2+z^2}}\right) = 0.$$

Чтобы решить equation на калькуляторе, let us represent его в виде $z = f(z)$:

$$f(z) = -\frac{901}{25} + \sqrt{28^2+(45+z)^2}\left(1 - \frac{224}{25\sqrt{28^2+z^2}}\right).$$

Погауается answer $z = 0$. Можно его подставить в $f(z)$□10□$z < -45$, составим следующее equation (меняется поле диска):

$$25\left(\frac{45+z}{\sqrt{28^2+(45+z)^2}} + 1\right) - 8\left(\frac{28}{\sqrt{28^2+(45+z)^2}} - \frac{28}{\sqrt{28^2+z^2}}\right) = 0.$$

Чтобы решить equation на калькуляторе, let us represent его в виде $z = f(z)$:

$$f(z) = -\frac{901}{25} - \sqrt{28^2+(45+z)^2}\left(1 + \frac{224}{25\sqrt{28^2+z^2}}\right).$$

Данное equation сходится очень медленно. Можно найти его корень методом бинарного поиска: substituting некоторое число and проводя итерации наблюдать, увеличивается or уменьшается quantity z. Правильный answer $z = -1632.6016357941451944046161090804794815785662664900$

Answer: \textbfAnswer:

$$\frac{z}{b} = 0.$$

$$\frac{z}{b} = -1632.60163579414519440461610908047948157856626649004333963591...$$

C3 **1.00 pts**

Under the conditions of the previous paragraph, the particle was placed in the equilibrium position closest to the cylinder, its mass m. Determine the angular frequency ω of small vibrations of the particle.

Select a point with coordinate $z = 0$. We expand the field strength and find that at this point $E(z) = -\alpha z$, $\alpha = 2\pi\sigma_0 k \cdot \frac{560}{2809}\frac{z}{b}$. From the vibration equation

$$m\ddot{z} = -q\alpha z$$

we get the answer.

Answer: \textbfAnswer:

$$\omega^2 = \frac{560\sigma_0}{5618\varepsilon_0 bm}$$